

Date: 20/11/2017 Roll No.
 B.E ECE THIRD SEMESTER
 END-SEMESTER EXAMINATION 2017, Nov.
 ECC08 SIGNALS AND SYSTEMS
 Time: 3:00 Hrs M.M.: 50

Question no. 1 is compulsory (2 X 5=10 Marks). Attempt any eight questions (each of 5 Marks) from the remaining. Assume missing data if any.

- Q[1] (a) Consider the waveform $x(t) = 4 + 2\cos 3t + 3\sin 4t$. Is it periodic? If yes, what is the fundamental time period T .
 (b) Consider the discrete time signal $x[n] = 1 - \sum_{k=3}^{\infty} \delta[n - 1 - k]$. Determine the values of the integers M and n_0 so that $x[n]$ may be expressed as $x[n] = u[Mn - n_0]$.
 (c) Prove that $\delta(2t) = \frac{1}{2}\delta(t)$.
 (d) Given that $x(t)$ is periodic with a time period T and fourier series coefficient X_k , compute the Fourier series coefficients of $y(t) = x(t)e^{j2\pi f_0 t} + x^*(t)e^{-j2\pi f_0 t}$, in terms of X_k .
 (e) Calculate the energy of the signal $y[n] = A\cos(\omega n + \phi)(u[n] - u[n - N])$.

- Q[2] Determine the response of the system with impulse response

$$h[n] = \left(\frac{1}{2}\right)^n u[n]$$

when the input is the complex exponential sequence

$$x[n] = Ae^{j\frac{\pi}{2}n}, \quad -\infty < n < \infty$$

- Q[3] Suppose we are given the following information about a signal $x(t)$:
 (a) $x(t)$ is real and odd.
 (b) $x(t)$ is periodic with period $T = 2$ and has Fourier coefficients X_k .
 (c) $X_k = 0$ for $|k| > 1$.
 (d) $\frac{1}{2} \int_0^2 |x(t)|^2 dt = 1$.
 Specify two different signals that satisfy these conditions.

- Q[4] Each of the two sequences $x[n]$ and $y[n]$ has a period $N = 4$, and the corresponding Fourier series coefficients are specified as $x[n] \leftrightarrow X_k$ and $y[n] \leftrightarrow Y_k$, where

$$X_0 = X_3 = \frac{1}{2}X_1 = \frac{1}{2}X_2 = 1, \text{ and } Y_0 = Y_1 = Y_2 = Y_3 = 1$$

Using the multiplication property, determine the Fourier series coefficients Z_k for the signal $z[n] = x[n]y[n]$.

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(94)

Q[5] An averager is defined by an input and output relationship

$$y(t) = \frac{1}{T_1 + T_2} \int_{t-T_1}^{t+T_2} x(\tau) d\tau$$

where T_1 and T_2 are constants. Is the system linear, time invariant, stable? Under what conditions is the system causal.

Q[6] The output $y(t)$ of a causal LTI system is related to the input $x(t)$ by the equation

$$\frac{dy(t)}{dt} + 10y(t) = \int_{-\infty}^{\infty} x(\tau)z(t-\tau)d\tau - x(t)$$

where $z(t) = e^{-t}u(t) + 3\delta(t)$. Find the frequency response $H(\omega)$ of this system. Also determine the impulse response of the system.

Q[7] Given that $x[n]$ has DTFT $X(e^{j\omega})$, express the Fourier transform of the following signals in terms of $X(e^{j\omega})$:

(a) $x_1[n] = \frac{x^*[n] + x[n]}{2}$.

(b) $x_2[n] = x[2n+1]$

(c) $x_3[n] = (n-1)^2 x[n]$.

Q[8] Consider a continuous time LTI system for which the input $x(t)$ and the output $y(t)$ are related by the differential equation

$$\frac{d^2 y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t)$$

(a) Determine the system function $H(s)$.

(b) Determine $h(t)$ for each of the following cases:

(i) The system is stable.

(ii) The system is causal.

Q[9] State and prove the properties of the ROC of Laplace-transform.

Q[10] When the input to an LTI system is

$$x[n] = \left(\frac{1}{3}\right)^n u[n] + (2)^n u[-n-1]$$

and the corresponding output is

$$y[n] = 5 \left(\frac{1}{3}\right)^n u[n] - 5 \left(\frac{2}{3}\right)^n u[n]$$

(a) Find the system function $H(z)$ of the system. Plot the poles and zeros of $H(z)$ and indicate the ROC.

(b) Find the impulse response $h[n]$ of the system.

(c) Is the system stable, causal.

Q[11] We are given the following facts about a discrete-time signal $x[n]$ with z-transform $X(z)$:

(a) $x[n]$ is real and right sided.

(b) $X(z)$ has exactly two poles.

(c) $X(z)$ has two zeroes at the origin.

(d) $X(z)$ has a pole at $z = (1/2)e^{j\pi/3}$.

(e) $X(1) = 8/3$.

Determine $X(z)$ and specify its ROC.

Q[12] Determine the pre-envelope and complex envelope of the following signal

$$x(t) = [A_c + m(t)]\cos(\omega_c t)$$

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